

Building a Real-Time Grocery Price Comparison System

This real-time grocery price comparison system has been designed for leading Indian e-commerce platforms. Built on microservices architecture, the system leverages FastAPI for backend services and Selenium for dynamic web scraping to deliver accurate, up-to-date pricing data.



Online grocery platforms have significantly reshaped consumer shopping habits in today's digital economy. With numerous e-commerce grocery services now available, users often struggle to compare prices efficiently across different platforms and make cost-effective decisions. This paper addresses that challenge by developing a system capable of providing real-time grocery price comparisons, allowing users to make more informed and economical purchasing choices.

This work sits at the intersection of consumer facing technology, web data extraction, and mobile application development. As grocery delivery platforms like Zepto, JioMart, Swiggy Instamart, and BigBasket continue to grow, price variations across these services have become frequent. However, the Indian market still lacks a centralised solution that offers real-time, user-friendly price comparison in a mobile-first format.

Our goal is to design and implement an Android application that fetches, compares, and displays current grocery prices from various vendors. The application is built to deliver a seamless user experience, supported by a scalable and modular backend. The backend system, developed using FastAPI and Selenium, handles real-time data scraping and exposes this information through RESTful APIs. For development and testing, the API is made accessible to the Android frontend using ngrok tunnelling. The frontend asynchronously retrieves pricing data and presents it through a clean, intuitive interface aligned with modern Android UI standards.

Key outcomes of this research work include a working mobile app that supports real-time grocery price comparisons, a modular backend supporting scraping and API services, and a flexible proof-of-concept that could be extended into intelligent shopping tools or dynamic pricing

analytics platforms in the future.

System architecture

The proposed system follows a modular, microservices-based architecture designed to aggregate grocery pricing data from multiple Indian e-commerce platforms in real-time. It emphasises scalability, low-latency API access, and smooth integration with mobile clients, while also supporting containerised deployment and CI/CD automation for streamlined development and updates. Figure 1 illustrates the high-level architecture.

Scraper microservices: Each grocery platform — Zepto, JioMart, Swiggy Instamart, BigBasket, and StarQuick — is associated with an independent scraping service. These services are implemented using Selenium WebDriver and operate in headless Chrome instances. Each scraper is encapsulated within a Docker container to maintain isolation, ease of deployment, and fault containment.